

claude-windows / dbc4cdae-04f6-4d06-af90-397bbce3fc57

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\dbc4cdae-04f6-4d06-af90-397bbce3fc57\subagents\agent-a497200f1395ac533.jsonl`
- SHA-256: `7908e750efc55acc6bc855ebf2b460b9dcc11fb2597249dd13ac927fde37c512`
- Source modified: `2026-07-06T19:37:32+00:00`
- Imported at: `2026-07-08T15:59:36+00:00`
- project: `subagents`
- session_id: `dbc4cdae-04f6-4d06-af90-397bbce3fc57`

Transcript

1. user (2026-07-06T19:34:30.257Z)

Research the WASB-SBDT repository (github.com/nttcom/WASB-SBDT) ? a shuttlecock/ball tracking "Widely Applicable Strong Baseline" from an ECCV 2022 / 2023 paper by Tarashima et al. (NTT Communications). I need falsifiable, citable facts about torch 2.x compatibility, licensing, and provenance.

Answer these with sources (URLs):

1. What is the LICENSE of the nttcom/WASB-SBDT repository (the code)? Is it MIT/Apache/BSD or something more restrictive?
2. What are the pretrained weights (e.g. `wasb_badminton_best.pth.tar`)? What dataset were the badminton weights trained on (e.g. the TrackNetV2 badminton dataset / "Shuttlecock" dataset)? Is there any stated license or usage restriction on the pretrained weights or that training dataset? This matters for commercial use.
3. What torch version does WASB-SBDT pin / require (check `requirements.txt`)? Does the repo use HRNet as its backbone/detector?
4. Known compatibility risks moving WASB code + weights from torch 1.11 to torch 2.x:
 - `torch.load weights_only` default changed to True in torch 2.6 ? would loading a `.pth.tar` checkpoint break?
 - Does the repo depend on `hydra-core` / `omegaconf` with version pins that conflict with modern setups?
 - Any removed/renamed torch APIs between 1.11 and 2.x that HRNet-style code commonly hits?
5. What benchmark and metric does the WASB paper report for badminton/shuttlecock? Get the actual accuracy/precision/recall/F1 numbers and the dataset name. Does it claim SOTA vs TrackNetV2?
6. The paper title, venue, year, and authors.

Use WebSearch for the WASB-SBDT GitHub repo, the paper (arXiv / ECCV / BMVC), and any papers-with-code entry. WebFetch the GitHub repo page, its LICENSE, `requirements.txt`, and the arXiv abstract/paper for exact quotes and numbers.

Return a concise bulleted list of findings, each with a source URL and a direct quote or specific fact. Flag uncertainties (especially around weights provenance/license) as open questions.

2. assistant (2026-07-06T19:34:34.804Z)

I'll research the WASB-SBDT repository systematically. Let me start by loading the web tools and running parallel searches.

3. assistant (2026-07-06T19:34:41.362Z)

I have the tools loaded. Let me run parallel searches to establish the repository, paper, and papers-with-code entries.

4. user (2026-07-06T19:34:48.625Z)

Web search results for query: "nttcom WASB-SBDT github shuttlecock tracking widely applicable strong baseline"

Links: [{"title": "Sports Ball Detection And Tracking", "url": "https://www.catalyzex.com/s/Sports%20Ball%20Detection%20And%20Tracking"}, {"title": "GitHub - nttcom/WASB-SBDT: [BMVC2023] Widely Applicable Strong Baseline for Sports Ball Detection and Tracking · GitHub", "url": "https://github.com/nttcom/WASB-SBDT"}, {"title": "(PDF) Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://www.researchgate.net/publication/376208967_Widely_Applicable_Strong_Baseline_for_Sports_Ball_Detection_and_Tracking"}, {"title": "WASB-SBDT/GET_STARTED.md at main · nttcom/WASB-SBDT", "url": "https://github.com/nttcom/WASB-SBDT/blob/main/GET_STARTED.md"}, {"title": "TrackNetV2: Efficient Shuttlecock Tracking Network | Request PDF", "url": "https://www.researchgate.net/publication/350935356_TrackNetV2_Efficient_Shuttlecock_Tracking_Network"}, {"title": "GitHub - qaz812345/TrackNetV3: Implementation of paper - TrackNetV3: Enhancing ShuttleCock Tracking with Augmentations and Trajectory Rectification · GitHub", "url": "https://github.com/qaz812345/TrackNetV3"}, {"title": "Ball Tracking for Tennis Video Annotation | Request PDF", "url": "https://www.researchgate.net/publication/287109346_Ball_Tracking_for_Tennis_Video_Annotation"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis | Request PDF", "url": "https://www.researchgate.net/publication/397895011_RacketVision_A_Multiple_Racket_Sports_Benchmark_for_Unified_Ball_and_Racket_Analysis"}, {"title": "NTT DOCOMO BUSINESS, Inc. · GitHub", "url": "https://github.com/nttcom"}, {"title": "TrackNetV2: Efficient Shuttlecock Tracking Network", "url": "https://paperswithcode.com/paper/tracknet2-efficient-shuttlecock-tracking"}]

Based on the search results, here's what I found about the WASB-SBDT project:

WASB-SBDT Overview

WASB (Widely Applicable Strong Baseline) is a Sports Ball Detection and Tracking (SBDT) baseline that can be applied to a wide range of sports categories including soccer, tennis, badminton, volleyball, and basketball.

Key Technical Approach

The approach is composed of (1) high-resolution feature extraction, (2) position-aware model training, and (3) inference considering temporal consistency, all of which are put together as a new SBDT baseline.

Research Validation

To validate the wide-applicability of the approach, the researchers compared their baseline with 6 state-of-the-art SBDT methods on 5 datasets from different sports categories. They achieved this by newly introducing two SBDT datasets, providing new ball annotations for two datasets, and re-implementing all the methods to ease extensive comparison. Experimental results demonstrate that their approach is substantially superior to existing methods on all the sports categories covered by the datasets.

Publication and Availability

Datasets and codes are available at <https://github.com/nttcom/WASB-SBDT>. The work was published at BMVC2023.

The repository includes support for training and evaluating the model on multiple sports including badminton (which involves shuttlecock tracking), tennis, basketball, volleyball, and soccer.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-07-06T19:34:50.775Z)

Web search results for query: "Tarashima WASB "Widely Applicable Strong Baseline" shuttlecock ball tracking paper ECCV BMVC 2023"

Links: [{"title": "A lightweight attention-guided multi-scale fusion network for real-time shuttlecock detection | Signal, Image and Video Processing | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s11760-025-05056-w"}, {"title": "TARASHIMA ET AL.: WIDELY APPLICABLE STRONG BASELINE FOR SBDT 1", "url": "https://papers.bmvc2023.org/0310.pdf"}, {"title": "Shuttlecock Detection Using Residual Learning in U-Net Architecture | Haq | JOIV : International Journal on Informatics Visualization", "url": "https://joiv.org/index.php/joiv/article/view/2132"}, {"title": "[2311.05237] Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://arxiv.org/abs/2311.05237"}, {"title": "GitHub - nttcom/WASB-SBDT: [BMVC2023] Widely Applicable Strong Baseline for Sports Ball Detection and Tracking · GitHub", "url": "https://github.com/nttcom/WASB-SBDT"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis", "url": "https://arxiv.org/html/2511.17045v3"}, {"title": "[2311.05237v1] Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://arxiv.org/abs/2311.05237v1"}, {"title": "(PDF) Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://www.researchgate.net/publication/376208967_Widely_Applicable_Strong_Baseline_for_Sports_Ball_Detection_and_Tracking"}, {"title": "Widely Applicable Strong Baseline for Sports Ball Detection and Tracking", "url": "https://proceedings.bmvc2023.org/310/"}, {"title": "TT3D: Table Tennis 3D Reconstruction", "url": "https://arxiv.org/html/2504.10035v2"}]

Based on the search results, I found comprehensive information about the paper you're looking for:

WASB: Widely Applicable Strong Baseline for Sports Ball Detection and Tracking

The paper by Tarashima et al., titled "A widely applicable strong baseline for sports ball detection and tracking (WASB)," was presented at the British Machine Vision Conference (BMVC) in 2023.

Paper Details

The full conference citation is: The 34th British Machine Vision Conference 2023 (BMVC 2023) held in Aberdeen, UK, November 20-24, 2023. The authors are Shuhei Tarashima, Muhammad Abdul Haq, Yushan Wang, and Norio Tagawa.

Key Contributions

The SBDT (Sports Ball Detection and Tracking) task can be uniformly defined through various ball-games, and wide applicability is an important property to be equipped by good SBDT methods.

The WASB achieves substantially better performance than 6 state-of-the-art (SOTA) SBDT methods on all the datasets, accomplished by introducing two novel SBDT datasets, providing two new manual annotations, and re-implementing all the SOTA methods.

Resources

The paper has been made available on arXiv (2311.05237), and code and datasets are provided for future SBDT research. The GitHub repository was released on 11/17/2023 and initially contained evaluation codes of pretrained WASB models.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-07-06T19:34:53.812Z)

I have the paper (arXiv 2311.05237, BMVC 2023) and repo. Now let me fetch the key primary sources in parallel: the GitHub repo page, LICENSE, requirements.txt, GET_STARTED.md, and the arXiv abstract.

7. user (2026-07-06T19:34:58.498Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

8. user (2026-07-06T19:34:59.672Z)

The server returned HTTP 404 Not Found.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

9. user (2026-07-06T19:35:01.144Z)

Extraction Results

(1) LICENSE Type

MIT license as stated in the repository header and footer.

(2) Pretrained Weights & Dataset Information

The page references a MODEL_ZOO.md file but does not display its contents. The only specific model mention is:

- General reference to "pretrained WASB models" being available
- News item states: "Repository is released. Now it contains evaluation codes of pretrained WASB models only."

No specific weight filenames, download links, or badminton-specific training dataset information are provided on this visible page content.

(3) License/Usage Restrictions

No statements about restrictions on pretrained weights or datasets appear on the visible page. The repository states an **MIT license**, which typically permits commercial use.

(4) Technical Dependencies

The page mentions:

- ***Tested with Python3.8, CUDA11.3***
- Reference to a Dockerfile for environment setup
- Links to GET_STARTED.md and MODEL_ZOO.md (contents not shown)

No explicit mentions of torch version, requirements.txt, hydra, omegaconf, or HRNet backbone are visible on this page excerpt.

Note: Detailed specifications likely exist in linked files (GET_STARTED.md, MODEL_ZOO.md) not displayed here.

10. user (2026-07-06T19:35:05.505Z)

Paper Information Extraction

Paper Title: "Widely Applicable Strong Baseline for Sports Ball Detection and Tracking"

Authors: Shuhei Tarashima, Muhammad Abdul Haq, Yushan Wang, Norio Tagawa

Venue: BMVC2023

Year: 2023

Abstract: The authors present a method for sports ball detection and tracking applicable across various sports. Their approach combines "high-resolution feature extraction, position-aware model training, and inference considering temporal consistency." They validated their baseline against six state-of-the-art methods across five datasets from different sports, introducing two new datasets and providing annotations for two existing ones. Results show their method achieved substantial improvements across all sports categories tested.

Specific Data Requested

****Badminton/Shuttlecock Results:**** Not found in the provided content.

****Badminton Dataset Name:**** Not specified in the provided content.

****TrackNetV2 Comparison Claim:**** No mention of outperforming TrackNetV2 appears in this excerpt.

****Note:**** The web page excerpt contains abstract and metadata only. Specific numeric results and dataset details would require accessing the full PDF paper.

11. assistant (2026-07-06T19:35:08.504Z)

The LICENSE and requirements.txt raw URLs 404'd ? likely the repo uses `master` branch or different filenames. Let me use the GitHub API to inspect the actual file tree and default branch, and fetch the BMVC PDF and MODEL_ZOO.md for weights/results.

12. user (2026-07-06T19:35:12.468Z)

default_branch: main
license: MIT / MIT License

13. user (2026-07-06T19:35:14.400Z)

Dockerfile
GET_STARTED.md
LICENSE.md
MODEL_ZOO.md
README.md
src/configs/dataloader/default.yaml
src/configs/dataset/badminton.yaml
src/configs/dataset/basketball.yaml
src/configs/dataset/soccer.yaml
src/configs/dataset/tennis.yaml
src/configs/dataset/volleyball.yaml
src/configs/detector/deepball.yaml
src/configs/detector/tracknetv2.yaml
src/configs/eval.yaml
src/configs/extract_frame.yaml
src/configs/loss/hm_bce.yaml
src/configs/loss/hm_combo.yaml
src/configs/loss/hm_focal.yaml
src/configs/loss/hm_mse.yaml
src/configs/loss/hm_qfl.yaml
src/configs/loss/hm_wbce.yaml
src/configs/loss/hm_wbce_s2.yaml
src/configs/loss/hm_wbce_s3.yaml
src/configs/loss/seg_ssd.yaml
src/configs/model/ballseg.yaml
src/configs/model/deepball.yaml
src/configs/model/deepball_large.yaml
src/configs/model/monotrack.yaml
src/configs/model/restracknetv2.yaml
src/configs/model/tracknetv2.yaml
src/configs/model/wasb.yaml
src/configs/optimizer/adadelta_constant.yaml
src/configs/optimizer/adadelta_multistep.yaml
src/configs/optimizer/adam_multistep.yaml
src/configs/runner/eval.yaml
src/configs/runner/extract_frame.yaml
src/configs/tracker/intra_frame_peak.yaml
src/configs/tracker/online.yaml

```
src/configs/transform/color.yaml
src/configs/transform/crop.yaml
src/configs/transform/default.yaml
src/configs/transform/full.yaml
src/configs/transform/hflip.yaml
```

14. assistant (2026-07-06T19:35:18.920Z)

License is confirmed MIT via GitHub API. The files are `LICENSE.md` and there's no `requirements.txt` in the tree ? deps are likely in the Dockerfile. There are Hydra-style YAML configs (confirming hydra/omegaconf usage). Let me fetch the key raw files.

15. user (2026-07-06T19:35:29.952Z)

```
===== LICENSE.md =====
```

MIT License

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The above copyright notice and this permission notice shall be included in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.

```
===== MODEL_ZOO.md =====
```

Model Zoo

We provide model weights of SOTA methods (including WASB) for each dataset.

We also provide a setup script to download all the listed models at once. See [here](https://github.com/starashima/WASB-SBDT_sandbox/blob/main/GET_STARTED.md#model-preparation) for more details.

```
|           | Soccer | Tennis | Badminton | Volleyball | Basketball |
| ----- | ----- | ---- | ---- | ---- | ---- |
| [DeepBall](https://arxiv.org/abs/1902.07304) | [Google
Drive](https://drive.google.com/file/d/1u-Y-RnIgu3H7bZCvv9JVQgkYsBRx0XS9/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1h5v5_kUw4c_4Uw7tpKSBSH4bJwlWm8ly/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1b-D2lrNuUNhZ90PaLSrIjvU86y00awFI/view?usp=drive_link) |
[Google Drive](https://drive.google.com/file/d/1mSwsewoBVAG-
EdyypkDk1lhIK915wGmo/view?usp=drive_link) | [Google
Drive](https://drive.google.com/file/d/1fUIVCzcKhfy5UiACF5T8m0pkr1x-Gu5C/view?usp=drive_link) |
| [DeepBall-Large](https://arxiv.org/abs/2311.05237) | [Google
Drive](https://drive.google.com/file/d/1-ak_vI8BiY9FoFVccxpYaJI_qJ6Rj0xw/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1hpCH6TM1Evowmu5JZAkthz141DLgxrgn/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1baHb1dtKHkXwQdGH8PB21hbdNV0u1dT1/view?usp=drive_link) |
[Google
```

Drive](https://drive.google.com/file/d/1tBLqbiokWpO6Xgh_fsNi-1kn0x6qh0H0/view?usp=drive_link) | [Google
Drive](https://drive.google.com/file/d/11ENjf5sAFkw6-jCGSXGfyU0t35axWdGH/view?usp=drive_link) |
| [BallSeg](https://arxiv.org/abs/2007.11876) | [Google
Drive](https://drive.google.com/file/d/1wbuxL-bKpG-0GVYa7BBewzmzCHKKIA15/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/14YEAcxwUUGvaL3km07fPSduJLBBNuk_D/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1zey7rEX8orvpT0c1qbD5FsQ93jfBmaML/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1FLJWvsupt5FXnPXMu41vPDWwIQBtU9d1/view?usp=drive_link) |
[Google Drive](https://drive.google.com/file/d/1dTV-
udhh04K0651yRd1tflhLMjh9uuBM/view?usp=drive_link) |
| [TrackNetV2](https://ieeexplore.ieee.org/document/9302757) | [Google
Drive](https://drive.google.com/file/d/1yTCC_8cWnMCL0pErk6UpepzQD9UF8Wpi/view?usp=drive_link) |
[Google Drive](https://drive.google.com/file/d/1BCnmvDX-LZpbkk4vLMEXMm-
uzCoqJzDx/view?usp=drive_link) | [Google
Drive](https://drive.google.com/file/d/1lCVYzua7jJfukQvWGypkYqr6PHA1EaPq/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/103j0dYp4k20avid4uyB9USCuwiphI4Kz/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1n-R_T0QyENSArYV8Qn8TneEroQkmYbT6/view?usp=drive_link) |
| [ResTrackNetV2](https://arxiv.org/abs/2311.05237) | [Google
Drive](https://drive.google.com/file/d/150_sbSm0XRCMDEJvsIeT7NFURJ_nucTB/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/112qHZpPWgqCeZbbFXC0oRulrVsokzWYZ/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1NUeuFp1xKxLzvGR5RHSFhrI3LvzKbaFo/view?usp=drive_link) |
[Google Drive](https://drive.google.com/file/d/1gnd-
VbUUGmiB_Obn7WHPmTuhd0zARN5c/view?usp=drive_link) | [Google
Drive](https://drive.google.com/file/d/1K0kxE8vdjnLZm7dzGvwVE-nDuzQWn0t9/view?usp=drive_link) |
| [MonoTrack](https://ieeexplore.ieee.org/document/9857202) | [Google
Drive](https://drive.google.com/file/d/1PBVMfjqLFiUN9M_4NkiAW0i_2alEk_vH/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1mc3yWf6ySlzF-1d-s_LNH1QWexHz2M3D/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1b7hDHU6q7HarB0CVSx46_Rurtn1J1Ko9/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/15dNX0oV_YiP7u2SbcyN5b2nNtMwVt3TT/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/1uM2FJLG11AtC0fHsur0qBBUuTehRJugs/view?usp=drive_link) |
| [WASB (Ours)](https://arxiv.org/abs/2311.05237) | [Google
Drive](https://drive.google.com/file/d/1pg0MpMtKZ6ziYEr4oyfKYP003hjLw94l/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/14AeyIOCQ2UaQmbZLNQJa1H_eSwxUXk7z/view?usp=drive_link) |
[Google
Drive](https://drive.google.com/file/d/17Ac0p05oryh1JwgtwTFQTjOKHY3umbDQu/view?usp=drive_link) |
[Google Drive](https://drive.google.com/file/d/1M9y4wPJqLc0K-z-
Bo5DP8Ft5XwJuLqIS/view?usp=drive_link) | [Google
Drive](https://drive.google.com/file/d/1nfECuSyJvPumz3njZCdFERSQQbERT8FU/view?usp=drive_link) |

===== GET_STARTED.md =====

Get Started

Data Preparation

Download [Soccer](https://pspagnolo.jimdo.free.com/download/),
[Tennis](https://nol.cs.nctu.edu.tw:234/open-source/TrackNet),
[Badminton](https://nol.cs.nctu.edu.tw:234/open-source/TrackNetv2),
[Volleyball](https://github.com/mostafa-saad/deep-activity-rec), and
[Basketball](https://ruiyan1995.github.io/SAM.html) and put them under ``<WASB-
SBDT_HOME>/datasets``.

- For Soccer, we provide a setup script. Run ``cd ./src && sh setup\_scripts/setup\_soccer.sh``.

- For Tennis, download ``Dataset.zip`` from [this link](https://nycu1-my.sharepoint.com/:u:/g/personal/tik_m365_nycu_edu_tw/ETCr6-M0e1VDhGcdMbvIjcsBu31AJT05xa_1cw8pHa7niA?e=55tLJ9) and put it under ``<WASB-SBDT\_HOME>`` directory. Then unzip the file and rename ``Dataset`` directory as ``tennis``.

- For Badminton, download ``TrackNetV2.zip`` from [this link](https://nycu1-my.sharepoint.com/:u:/g/personal/tik_m365_nycu_edu_tw/EWisYhAiai9Ju7L-tQp0yKEBZJd9VQkKqsFrjcqqYIDP-g?e=S0AB1Z) and put it under ``<WASB-SBDT\_HOME>`` directory. Then use a setup script by running ``cd ./src && sh setup\_scripts/setup\_badminton.sh``.

- For Volleyball, download ``volleyball\_.zip`` from [this link](https://drive.google.com/drive/folders/1rmsrG1mgkwX0Khsr-QYoi9Ss92wQmCOS?usp=sharing) and ``volleyball\_ball\_annotations.zip`` from [this link](https://drive.google.com/file/d/1urZpZiiepC85JD1u3VeURgUpztRgI0yl/edit) and unzip them. Then put the resulting directories as follows.

- For Basketball, download all the zip segments shown [here](https://ruiyan1995.github.io/SAM.html) and unzip them to generate ``NBA\_data``. Then run a setup script i.e., ``cd ./src && sh setup\_scripts/setup\_basketball.sh``.

Data structure should be the following:

```

...
  datasets
  |-----soccer
  |       ?-----videos
  |       ?-----frames
  |       ?-----annos
  ?-----tennis /* renamed from Dataset */
  |       ?-----game1
  |       ?-----...
  |       ?-----game10
  ?-----badminton
  |       ?-----match1
  |       ?-----...
  |       ?-----match26
  |       ?-----test_match1
  |       ?-----...
  |       ?-----test_match3
  ?-----volleyball
  |       ?-----videos
  |       ?-----volleyball_ball_annotation
  ?-----basketball /* renamed from NBA_data */
  |       ?-----videos
  |       ?-----ball-annos
  |
  src
...

```

Model Preparation

Download pretrained models listed in [MODEL_ZOO.md](./MODEL_ZOO.md).

We also provide a setup script to download all the listed models at once. Run ``cd ./src && sh setup\_scripts/setup\_weights.sh``, then models are located in ``<WASB-SBDT\_HOME>/pretrained\_weights``.

Evaluation

Here we show the evaluation commands to reproduce the results of Table 2 and Table 3 in [our paper](https://arxiv.org/abs/2311.05237).

[WASB (Ours, Step=3)](https://arxiv.org/abs/2311.05237)

...

Soccer

```
python3 main.py --config-name=eval dataset=soccer model=wasb
detector.model_path=../pretrained_weights/wasb_soccer_best.pth.tar
```

Tennis

```
python3 main.py --config-name=eval dataset=tennis model=wasb
detector.model_path=../pretrained_weights/wasb_tennis_best.pth.tar
```

Badminton

```
python3 main.py --config-name=eval dataset=badminton model=wasb
detector.model_path=../pretrained_weights/wasb_badminton_best.pth.tar
```

Volleyball

```
python3 main.py --config-name=eval dataset=volleyball model=wasb
detector.model_path=../pretrained_weights/wasb_volleyball_best.pth.tar
```

Basketball

```
python3 main.py --config-name=eval dataset=basketball model=wasb
detector.model_path=../pretrained_weights/wasb_basketball_best.pth.tar
```
```

**[WASB (Ours, Step=1)](<https://arxiv.org/abs/2311.05237>)**

```

Soccer

```
python3 main.py --config-name=eval dataset=soccer model=wasb
detector.model_path=../pretrained_weights/wasb_soccer_best.pth.tar detector.step=1
```

Tennis

```
python3 main.py --config-name=eval dataset=tennis model=wasb
detector.model_path=../pretrained_weights/wasb_tennis_best.pth.tar detector.step=1
```

Badminton

```
python3 main.py --config-name=eval dataset=badminton model=wasb
detector.model_path=../pretrained_weights/wasb_badminton_best.pth.tar detector.step=1
```

Volleyball

```
python3 main.py --config-name=eval dataset=volleyball model=wasb
detector.model_path=../pretrained_weights/wasb_volleyball_best.pth.tar detector.step=1
```

Basketball

```
python3 main.py --config-name=eval dataset=basketball model=wasb
detector.model_path=../pretrained_weights/wasb_basketball_best.pth.tar detector.step=1
```
```

**[MonoTrack [CVPRW2022]](<https://ieeexplore.ieee.org/document/9857202>)**

```

Soccer

```
python3 main.py --config-name=eval dataset=soccer model=monotrack
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/monotrack_soccer_best.pth.tar tracker=intra_frame_peak
```

Tennis

```
python3 main.py --config-name=eval dataset=tennis model=monotrack
detector.postprocessor.use_hm_weight=False
```

```
detector.model_path=../pretrained_weights/monotrack_tennis_best.pth.tar tracker=intra_frame_peak
```

Badminton

```
python3 main.py --config-name=eval dataset=badminton model=monotrack
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/monotrack_badminton_best.pth.tar
tracker=intra_frame_peak
```

Volleyball

```
python3 main.py --config-name=eval dataset=volleyball model=monotrack
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/monotrack_volleyball_best.pth.tar
tracker=intra_frame_peak
```

Basketball

```
python3 main.py --config-name=eval dataset=basketball model=monotrack
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/monotrack_basketball_best.pth.tar
tracker=intra_frame_peak
...
```

[ResTrackNetV2](<https://arxiv.org/abs/2311.05237>)

...

Soccer

```
python3 main.py --config-name=eval dataset=soccer model=restracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/restracknetv2_soccer_best.pth.tar
tracker=intra_frame_peak
```

Tennis

```
python3 main.py --config-name=eval dataset=tennis model=restracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/restracknetv2_tennis_best.pth.tar
tracker=intra_frame_peak
```

Badminton

```
python3 main.py --config-name=eval dataset=badminton model=restracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/restracknetv2_badminton_best.pth.tar
tracker=intra_frame_peak
```

Volleyball

```
python3 main.py --config-name=eval dataset=volleyball model=restracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/restracknetv2_volleyball_best.pth.tar
tracker=intra_frame_peak
```

Basketball

```
python3 main.py --config-name=eval dataset=basketball model=restracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/restracknetv2_basketball_best.pth.tar
tracker=intra_frame_peak
...
```

[TrackNetV2 [ICPAI2020]](<https://ieeexplore.ieee.org/document/9302757>)

...

Soccer

```
python3 main.py --config-name=eval dataset=soccer model=tracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=./pretrained_weights/tracknetv2_soccer_best.pth.tar
tracker=intra_frame_peak
```

Tennis

```
python3 main.py --config-name=eval dataset=tennis model=tracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=./pretrained_weights/tracknetv2_tennis_best.pth.tar
tracker=intra_frame_peak
```

Badminton

```
python3 main.py --config-name=eval dataset=badminton model=tracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=./pretrained_weights/tracknetv2_badminton_best.pth.tar
tracker=intra_frame_peak
```

Volleyball

```
python3 main.py --config-name=eval dataset=volleyball model=tracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=./pretrained_weights/tracknetv2_volleyball_best.pth.tar
tracker=intra_frame_peak
```

Basketball

```
python3 main.py --config-name=eval dataset=basketball model=tracknetv2
detector.postprocessor.use_hm_weight=False
detector.model_path=./pretrained_weights/tracknetv2_basketball_best.pth.tar
tracker=intra_frame_peak
...
```

[BallSeg [MMSports2019]](<https://arxiv.org/abs/2007.11876>)

...

Soccer

```
python3 main.py --config-name=eval dataset=soccer model=ballseg detector.step=1
detector.postprocessor.use_hm_weight=False
detector.model_path=./pretrained_weights/ballseg_soccer_best.pth.tar tracker=intra_frame_peak
```

Tennis

```
python3 main.py --config-name=eval dataset=tennis model=ballseg detector.step=1
detector.postprocessor.use_hm_weight=False
detector.model_path=./pretrained_weights/ballseg_tennis_best.pth.tar tracker=intra_frame_peak
```

Badminton

```
python3 main.py --config-name=eval dataset=badminton model=ballseg detector.step=1
detector.postprocessor.use_hm_weight=False
detector.model_path=./pretrained_weights/ballseg_badminton_best.pth.tar
tracker=intra_frame_peak
```

Volleyball

```
python3 main.py --config-name=eval dataset=volleyball model=ballseg detector.step=1
detector.postprocessor.use_hm_weight=False
detector.model_path=./pretrained_weights/ballseg_volleyball_best.pth.tar
tracker=intra_frame_peak
```

Basketball

```
python3 main.py --config-name=eval dataset=basketball model=ballseg detector.step=1
```

```
detector.postprocessor.use_hm_weight=False
detector.model_path=../pretrained_weights/ballseg_basketball_best.pth.tar
tracker=intra_frame_peak
...
```

[DeepBall [VISAPP2019]](<https://arxiv.org/abs/1902.07304>)

...

Soccer

```
python3 main.py --config-name=eval dataset=soccer model=deepball detector=deepball
detector.model_path=../pretrained_weights/deepball_soccer_best.pth.tar detector.step=1
tracker=intra_frame_peak
```

Tennis

```
python3 main.py --config-name=eval dataset=tennis model=deepball detector=deepball
detector.model_path=../pretrained_weights/deepball_tennis_best.pth.tar detector.step=1
tracker=intra_frame_peak
```

Badminton

```
python3 main.py --config-name=eval dataset=badminton model=deepball detector=deepball
detector.model_path=../pretrained_weights/deepball_badminton_best.pth.tar detector.step=1
tracker=intra_frame_peak
```

Volleyball

```
python3 main.py --config-name=eval dataset=volleyball model=deepball detector=deepball
detector.model_path=../pretrained_weights/deepball_volleyball_best.pth.tar detector.step=1
tracker=intra_frame_peak
```

Basketball

```
python3 main.py --config-name=eval dataset=basketball model=deepball detector=deepball
detector.model_path=../pretrained_weights/deepball_basketball_best.pth.tar detector.step=1
tracker=intra_frame_peak
...
```

[DeepBall-Large](<https://arxiv.org/abs/2311.05237>)

...

Soccer

```
python3 main.py --config-name=eval dataset=soccer model=deepball_large detector=deepball
detector.model_path=../pretrained_weights/deepball-large_soccer_best.pth.tar detector.step=1
tracker=intra_frame_peak
```

Tennis

```
python3 main.py --config-name=eval dataset=tennis model=deepball_large detector=deepball
detector.model_path=../pretrained_weights/deepball-large_tennis_best.pth.tar detector.step=1
tracker=intra_frame_peak
```

Badminton

```
python3 main.py --config-name=eval dataset=badminton model=deepball_large detector=deepball
detector.model_path=../pretrained_weights/deepball-large_badminton_best.pth.tar detector.step=1
tracker=intra_frame_peak
```

Volleyball

```
python3 main.py --config-name=eval dataset=volleyball model=deepball_large detector=deepball
detector.model_path=../pretrained_weights/deepball-large_volleyball_best.pth.tar detector.step=1
tracker=intra_frame_peak
```

Basketball

```
python3 main.py --config-name=eval dataset=basketball model=deepball_large detector=deepball
detector.model_path=./pretrained_weights/deepball-large_basketball_best.pth.tar detector.step=1
tracker=intra_frame_peak
...
```

Training

TBA

```
===== Dockerfile =====
FROM nvcr.io/nvidia/tensorrt:21.06-py3

MAINTAINER shuheitarashima@ntt.com
LABEL OBJECT="Dockerfile for Widely Applicable Strong Baseline for Sports Ball Detection and
Tracking [BMVC2023]"
ARG http_proxy
ARG https_proxy
ENV http_proxy ${http_proxy}
ENV https_proxy ${https_proxy}
ENV DEBIAN_FRONTEND noninteractive

RUN apt-get update && apt-get install -y --no-install-recommends\
    apt-utils\
    apt-transport-https\
    ca-certificates\
    software-properties-common\
&& apt-get clean\
&& rm -rf /var/lib/apt/lists/*
RUN apt-get update && apt-get install -y \
    git\
    vim\
    cmake\
    libsm6\
    libxext6\
    libxrender-dev\
    libgl1-mesa-glx\
    locales\
    build-essential\
    zlib1g-dev\
    libssl-dev\
    libbz2-dev\
    libreadline-dev\
    libsqlite3-dev\
    libhdf5-dev\
    libncursesw5-dev\
    libgdbm-dev\
    liblzma-dev\
    libdb-dev\
    libffi-dev\
    uuid-dev\
    tk-dev\
&& apt-get clean\
&& rm -rf /var/lib/apt/lists/*

ENV LANG en_US.UTF-8
ENV LANGUAGE en_US:en
ENV LC_ALL en_US.UTF-8
RUN locale-gen en_US.UTF-8

RUN pip3 install numpy==1.22.4 torch==1.11.0+cu113 torchvision==0.12.0+cu113 --extra-index-url
https://download.pytorch.org/whl/cu113
RUN pip3 install hydra-core==1.2.0 tqdm==4.64.0 opencv-python==4.6.0.66 scikit-learn==1.1.1
scikit-image==0.19.3 pandas==1.3.5 einops==0.4.1 timm==0.6.5 matplotlib==3.5.2
```

```
COPY src /root/src
WORKDIR /root/src
#COPY .vimrc /root/
```

===== README.md =====

WASB: Widely Applicable Strong Baseline for Sports Ball Detection and Tracking

Code & dataset repository for the paper: ****[Widely Applicable Strong Baseline for Sports Ball Detection and Tracking](https://arxiv.org/abs/2311.05237)****

Shuhei Tarashima, Muhammad Abdul Haq, Yushan Wang, Norio Tagawa

```
[![arXiv](https://img.shields.io/badge/arXiv-2311.05237-00ff00.svg)](https://arxiv.org/abs/2311.05237) [![License: MIT](https://img.shields.io/badge/License-MIT-yellow.svg)](https://opensource.org/licenses/MIT)
![test](https://img.shields.io/static/v1?label=By&message=Pytorch&color=red)
```

We present Widely Applicable Strong Baseline (WASB), a Sports Ball Detection and Tracking (SBDT) baseline that can be applied to wide range of sports categories :soccer: :tennis: :badminton: :volleyball: :basketball: .

<https://github.com/nttcom/WASB-SBDT/assets/63090948/8889ef53-62c7-4c97-9b33-8bf386489ba1>

News

- [11/23/2023] [Our BMVC2023 proceeding](https://proceedings.bmvc2023.org/310/) is available! Thank you, BMVC2023 organizers!
- [11/23/2023] Evaluation codes of DeepBall, DeepBall-Large and BallSeg are added!
- [11/21/2023] Evaluation codes of TrackNetV2, ResTrackNetV2 and MonoTrack are added!
- [11/17/2023] Repository is released. Now it contains evaluation codes of pretrained WASB models only. Other models will be coming soon!
- [11/09/2023] Our [arXiv preprint](https://arxiv.org/abs/2311.05237) is released.

Installation and Setup

Tested with Python3.8, CUDA11.3 on Ubuntu 18.04 (4 V100 GPUs inside). We recommend to use the [Dockerfile](./Dockerfile) provided in this repo (with `--it` option when running the container).

- See [GET_STARTED.md](./GET_STARTED.md) for how to get started with SBDT models.
- See [MODEL_ZOO.md](./MODEL_ZOO.md) for available model weights.

Citation

If you find this work useful, please consider to cite our paper:

...

```
@inproceedings{tarashima2023wasb,
  title={Widely Applicable Strong Baseline for Sports Ball Detection and Tracking},
  author={Tarashima, Shuhei and Haq, Muhammad Abdul and Wang, Yushan and Tagawa, Norio},
  booktitle={BMVC},
  year={2023}
}
```

===== src/configs/model/wasb.yaml =====

```
name: hrnet
frames_in: 3
frames_out: 3
inp_height: 288
```

```

inp_width: 512
out_height: 288
out_width: 512
rgb_diff: False
out_scales: [0]
MODEL:
  EXTRA:
    FINAL_CONV_KERNEL: 1
    PRETRAINED_LAYERS: ['*']
  STEM:
    INPLANES: 64
    STRIDES: [1,1]
  STAGE1:
    NUM_MODULES: 1
    NUM_BRANCHES: 1
    BLOCK: BOTTLENECK
    NUM_BLOCKS: [1]
    NUM_CHANNELS: [32]
    FUSE_METHOD: SUM
  STAGE2:
    NUM_MODULES: 1
    NUM_BRANCHES: 2
    BLOCK: BASIC
    NUM_BLOCKS: [2,2]
    NUM_CHANNELS: [16,32]
    FUSE_METHOD: SUM
  STAGE3:
    NUM_MODULES: 1
    NUM_BRANCHES: 3
    BLOCK: BASIC
    NUM_BLOCKS: [2,2,2]
    NUM_CHANNELS: [16,32,64]
    FUSE_METHOD: SUM
  STAGE4:
    NUM_MODULES: 1
    NUM_BRANCHES: 4
    BLOCK: BASIC
    NUM_BLOCKS: [2,2,2,2]
    NUM_CHANNELS: [16,32,64,128]
    FUSE_METHOD: SUM
  DECONV:
    NUM_DECONVS: 0
    KERNEL_SIZE: []
    NUM_BASIC_BLOCKS: 2
  INIT_WEIGHTS: True

```

===== src/configs/dataset/badminton.yaml =====

```

name: badminton
root_dir: /root/datasets/badminton
video_dirname: video
frame_dirname: frame
csv_dirname: csv
train:
  matches : [ 'match1',
              'match2',
              'match3',
              'match4',
              'match5',
              'match6',
              'match7',
              'match8',
              'match9',
              'match10',
              'match11',

```

```

        'match12',
        'match13',
        'match14',
        'match15',
        'match16',
        'match17',
        'match18',
        'match19',
        'match20',
        'match21',
        'match22',
        'match23',
        'match24',
        'match25',
        'match26']
    num_clip_ratio: 1.0
test:
    matches: ['test_match1',
              'test_match2',
              'test_match3']
    num_clip_ratio: 1.0

```

16. assistant (2026-07-06T19:35:34.785Z)

Excellent ? I now have nearly everything from primary sources. Key findings: torch 1.11.0+cu113 pinned in Dockerfile, hydra-core==1.2.0, backbone is `name: hrnet` (confirmed HRNet), badminton dataset is TrackNetV2 (NCTU download link), weights are `.pth.tar` on Google Drive. Now I need the actual benchmark numbers from the BMVC paper. Let me fetch the BMVC PDF and check how the code loads checkpoints (torch.load / weights_only relevance) and whether HRNet code uses any deprecated APIs.

17. user (2026-07-06T19:35:42.286Z)

I cannot extract the requested badminton/shuttlecock tracking results from the provided content. The text appears to be raw PDF binary data containing embedded fonts and compressed streams rather than readable document content.

To obtain the specific numeric results you're seeking regarding WASB versus TrackNetV2 performance metrics on badminton datasets, you would need to:

1. Provide the document in a text-readable format (plain text, properly extracted PDF text, or HTML)
2. Specify the exact paper title or authors
3. Check the original publication directly

The current file format prevents accurate extraction of tables, metrics, and comparative performance claims.

[Binary content (application/pdf, 1.1MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\dbc4cdae-04f6-4d06-af90-397bbce3fc57\tool-results\webfetch-1783366538578-mlblhd.pdf]

18. user (2026-07-06T19:35:43.620Z)

```

src/dataloaders/__init__.py
src/dataloaders/dataset_loader.py
src/dataloaders/heatmaps/__init__.py
src/dataloaders/heatmaps/heatmaps.py
src/dataloaders/img_transforms.py
src/dataloaders/samplers/__init__.py
src/dataloaders/samplers/samplers.py
src/dataloaders/seq_transforms.py
src/detectors/__init__.py
src/detectors/deepball_detector.py

```



```

src/detectors/deepball_postprocessor.py
src/detectors/detector.py
src/detectors/postprocessor.py
src/main.py
src/models/hrnet.py
src/runners/runner_utils.py
src/utils/__init__.py
src/utils/dataclasses.py
src/utils/evaluator.py
src/utils/file.py
src/utils/heatmap.py
src/utils/image.py
src/utils/refine_gt.py
src/utils/transforms.py
src/vis.py

```

19. assistant (2026-07-06T19:35:53.454Z)

The BMVC PDF was saved locally as a real PDF file. Let me read it directly, and also fetch the checkpoint-loading code (`torch.load` usage) and the HRNet model file to assess torch 2.x compatibility.

20. user (2026-07-06T19:35:54.985Z)

pdftoppm is not installed. Install poppler-utils (e.g. `brew install poppler` or `apt-get install poppler-utils`) to enable PDF page rendering.

21. user (2026-07-06T19:35:59.667Z)

```

===== src/detectors/detector.py =====
5:import numpy as np
45:     if not torch.cuda.is_available():
58:         checkpoint = torch.load(model_path)
59:         self._model.load_state_dict(checkpoint['model_state_dict'])
===== src/utils/transforms.py =====
7:import numpy as np
16:     X2 = np.sum(X**2, axis=1) # shape of (m)
17:     Y2 = np.sum(Y**2, axis=1) # shape of (n)
18:     XY = np.matmul(X, Y.T)
20:     return np.sqrt(X2 - 2*XY + Y2)
67:     torch.save(state, model_path)
76:     np.random.seed(seed)
77:     torch.manual_seed(seed)
78:     torch.cuda.manual_seed(seed)
79:     torch.cuda.manual_seed_all(seed)
80:     torch.backends.cudnn.benchmark = False
81:     torch.backends.cudnn.deterministic = True
84:     y = torch.clamp(x.sigmoid_(), min=1e-4, max=1-1e-4)
97:     topk_scores, topk_inds = torch.topk(scores.view(batch, seq, -1), 1)
===== src/utils/file.py =====
3:import numpy as np
32:     if np.isnan(x) or np.isnan(y):

```

22. assistant (2026-07-06T19:36:05.700Z)

Critical finding confirmed: `detector.py:58` uses `checkpoint = torch.load(model\_path)` with ****no `weights\_only` argument**** and loads `checkpoint['model\_state\_dict']` ? this is the torch 2.6 `weights\_only=True` default risk. Let me read the full detector.py loading block and the HRNet model file for deprecated APIs, and extract the PDF text via a Python approach.

23. user (2026-07-06T19:36:11.755Z)

```

===== detector.py lines 40-75 =====
_, self._transform = build_img_transforms(cfg)

```

```

self._device = cfg['runner']['device']
if self._device!='cuda':
    assert 0, 'device=cpu not supported'
if not torch.cuda.is_available():
    assert 0, 'GPU NOT available'
self._gpus = cfg['runner']['gpus']

if model is None:
    self._model = build_model(cfg)
    model_path = cfg['detector']['model_path']
    if model_path is None:
        output_dir = HydraConfig.get().run.dir
        model_path = osp.join(output_dir, 'best_model.pth.tar')
        log.info('Checkpoint is not specified, so it is set as the best model in
{}'.format(output_dir))
        if not osp.exists(model_path):
            FileNotFoundError('{} not found'.format(model_path))
        checkpoint = torch.load(model_path)
        self._model.load_state_dict(checkpoint['model_state_dict'])
        self._model = self._model.to(self._device)
        self._model = nn.DataParallel(self._model, device_ids=self._gpus)
    else:
        self._model = model

self._model.eval()

postprocessor_name = cfg['detector']['postprocessor']['name']
if not postprocessor_name in self.__postprocessor_factory.keys():
    raise KeyError('invalid dataset: {}'.format(postprocessor_name ))
self._postprocessor = self.__postprocessor_factory[postprocessor_name](cfg)

@property
def frames_in(self):
    return self._frames_in

```

===== hrnet.py deprecated-API scan =====

```

19:import torch
20:import torch.nn as nn
39:    self.bn1 = nn.BatchNorm2d(planes, momentum=BN_MOMENTUM)
42:    self.bn2 = nn.BatchNorm2d(planes, momentum=BN_MOMENTUM)
71:    self.bn1 = nn.BatchNorm2d(planes, momentum=BN_MOMENTUM)
74:    self.bn2 = nn.BatchNorm2d(planes, momentum=BN_MOMENTUM)
77:    self.bn3 = nn.BatchNorm2d(planes * self.expansion,
78:                             momentum=BN_MOMENTUM)
153:        nn.BatchNorm2d(num_channels[branch_index] * block.expansion,
154:                        momentum=BN_MOMENTUM),
195:        nn.BatchNorm2d(num_inchannels[i]),
208:        nn.BatchNorm2d(num_outchannels_conv3x3))
215:        nn.BatchNorm2d(num_outchannels_conv3x3),
265:    self.bn1 = nn.BatchNorm2d(self._stem_inplanes, momentum=BN_MOMENTUM)
267:    self.bn2 = nn.BatchNorm2d(self._stem_inplanes, momentum=BN_MOMENTUM)
352:        nn.BatchNorm2d(output_channels, momentum=BN_MOMENTUM),
376:        nn.BatchNorm2d(
377:            num_channels_cur_layer[i], momentum=BN_MOMENTUM),
390:        nn.BatchNorm2d(outchannels, momentum=BN_MOMENTUM),
402:        nn.BatchNorm2d(planes * block.expansion, momentum=BN_MOMENTUM),
491:    elif isinstance(m, nn.BatchNorm2d):
495:        pretrained_dict = torch.load(pretrained)
511:    import torch

```

24. user (2026-07-06T19:36:17.377Z)

sports such as badminton [83], baseball [74], basketball [24], golf [30], soccer [72, 78], tennis [64], table tennis [18], and volleyball [15]. Several commercial systems like Hawk-

Figure 1: Exemplar ball trajectories extracted from soccer, tennis, badminton, volleyball and specific sports (e.g., badminton [9], baseball [74], basketball [5, 6, 7, 8, 11], golf [53, 54], 59, 64, 69, 77, 85, 86, 88, 89, 90, 97, 101, 103, 109, 110], table tennis [13, 18, 23, 26, 58], category, we use 5 datasets from different sports categories (i.e., badminton, basketball be fed into the model to capture ball dynamics. TrackNet and its variants [32, 50, 75] are resolution and semantically-rich feature representation to precisely detect tiny sports balls. (d), which results in more precise localization as shown in (e).

be clues to precisely estimate a ball position. We propose to compute a ball position as the Basketball and Soccer are newly annotated by us. Statistics are summarized in Table 1. Tennis [32]. This dataset was introduced along with the TrackNet work [32], but was not Badminton [75]. This dataset was introduced by the TrackNetV2 work [75]. Following the Badminton [75]1280×720 30 26 172 78558 11 .8±12.2 3 29 12656 12 .5±12.9

Table 1: Summary of 5 SBDT datasets used in our evaluation. Among them, V olleyball and novel frame-wise manual annotations of 2D ball position. In this table, ?resolution? represent largest among the five datasets (cf. Table 1). Also, camera frequently moves and zooms in BallSeg [80], TrackNetV2 [75] and MonoTrack [50]. For DeepBall, since its original deployed an unpublished variant9 of TrackNetV2, where residual connection and transposed convolution are additionally employed. We call this variant as ResTrackNetV2.

We report the performances of our SOTA re-implementations in Table 2. It shows that the accuracy of our TrackNetV2 [75] implementation on the Badminton dataset is 85.6, while Table IV in [75] shows that the original implementation scores 85.2, which indicates the 7https://nol.cs.nctu.edu.tw:234/open-source/TrackNetV2

8https://nol.cs.nctu.edu.tw:234/open-source/TrackNet

9https://github.com/Chang-Chia-Chi/TrackNet-Badminton-Tracking-tensorflow2

Soccer Tennis Badminton V olleyball Basketball# param.F1? Acc.? AP? FPS? F1 Acc. AP FPSF1 Acc. AP FPSF1 Acc. AP FPS

DeepBall [40, 41] 0.1M44.5 92.7 26.3 44.647.4 32.3 47.0 52.152.4 38.6 60.0 57.164.4 50.7 49.2 21.10.0 12.9 0.0 30.3DeepBall-Large 1.0M44.9 89.5 34.0 42.046.7 31.6 35.1 47.750.6 36.8 59.5 53.070.4 57.5

W ASB (Ours,

Step=3)1.5M88.397.983.655.794.089.091.058.291.687.088.570.486.577.979.918.080.671.371.530.2W ASB (Ours, Step=1)1.5M88.297.986.223.695.691.894.235.293.189.091.634.388.080.083.215.882.673.4

Table 2: Benchmark results of SBDT methods on 5 SBDT datasets. We set the distance threshold ? = 4 [pixel] to compute F1, Accuracy (Acc.) and Average Precision (AP), all of correctness (or, superiority) of our TrackNetV2 implementation. Unfortunately, such a validation makes the validation intractable. For BallSeg [80], neither its specific architecture is pre-We evaluate SBDT models using F1, Accuracy (Acc.) and Average Precision (AP). With a false positive or false negative. F1 and Acc. can be directly computed with the results, while Following TrackNetV2 and its variants [50, 75], N (cf. §3.1) is set to 3 and each image is 100F1

TrackNetV2

ResTrackNetV2

Badminton

100Average Precision (AP)

Figure 6: F1 (first row), Accuracy (second row) and Average Precision (third row) of SBDT

Table 2 shows the benchmark results of SBDT methods on our datasets, using the fixed

Figure 6 shows F1, Accuracy and AP scores of SBDT methods with different distance

depend on the dataset, while TrackNetV2 [75], ResTrackNetV2 and MonoTrack [50] stably

Table 3 shows the ablation results with respect to the model design discussed in §3.1. As

Table 4 represents the ablation results to evaluate the techniques introduced in §3.2 and

Soccer Tennis Badminton V olleyball Basketball# param.F1? Acc.? AP? FPS? F1 Acc. AP FPSF1 Acc. AP FPSF1 Acc. AP FPS

Figure 3 (a) 1.5M81.7 96.9 71.7 85.785.5 75.4 75.6 56.786.8 80.4 80.3 77.184.3 74.7 77.0

17.677.4 67.3 67.1 30.8Figure 3 (b) 1.5M86.4 97.6 79.0 76.791.9 85.4 86.7 60.390.5 85.5 86.2 76.285.0 75.8 77.2

Table 3: Ablations with respect to the model design (cf. §3.2). Notice that in all the cases Soccer Tennis BadmintonHLSM (§3.2) CoH (§3.3) Online Tracking (§3.3) Step=1 (§3.3)F1? Acc.? AP? F1 Acc. AP F1 Acc. AP

(The best scores of existing methods (cf.Table 2)) 85.2 97.7 78.6 92.1 85.9 87.390.9 85.9 84.9 87.3 97.7 80.1 93.1 88.1 88.591.1 86.3 85.5

! 87.8 97.8 81.1 93.7 88.6 89.491.4 86.6 86.2

! ! 88.3 97.9 83.6 93.9 88.8 90.891.6 87.0 88.5

! ! ! 88.3 97.9 83.6 94.0 89.0 91.091.6 87.0 88.5

! ! ! ! 88.2 97.9 86.2 95.6 91.8 94.293.1 89.0 91.6

Table 4: Ablation results with respect to our proposed training (cf. §3.2) and inference (cf.

§3.3) schemes on the Soccer, Tennis and Badminton datasets.

with a few exceptions (e.g., online tracking does not contribute on the Soccer and Badminton of existing methods (cf. first row in Table 4). This indicates the superiority of our high-Data of a Badminton Match Based on 2-D Seriate Images. Tsinghua Science and Ranking of Broadcasting Table Tennis Video. InThe Proceedings of the Multiconfer-TrackNet: A Deep Learning Network for Tracking High-speed and Tiny Objects in Monocular Badminton Video. In Proceedings of the IEEE/CVF Conference on Com-[58] Hnin Myint, Patrick Wong, Laurence Dooley, and Adrian Hopgood. Tracking a Table 1999.

Chung, and Tsi-Uí ?Ik. TrackNetV2: Efficient Shuttlecock Tracking Network. In 2020 poral and Spatial Video Analysis of Table Tennis. In Proceedings of the IEEE/CVF casting in Badminton. Proceedings of the AAAI Conference on Artificial Intelligence, [86] Kam Cheung Patrik Wong and Laurence S. Dooley. High-motion Table Tennis Ball Trajectory for Table Tennis Robot. IEEE Transactions on Instrumentation and Mea-

25. assistant (2026-07-06T19:36:21.650Z)

I now have the exact benchmark numbers and all technical details. Let me parse the badminton column from Table 2 carefully. The full TrackNetV2 badminton row is needed ? let me extract it precisely from the PDF, plus confirm the HRNet `torch.load(pretrained)` at line 495 (only used when INIT_WEIGHTS loads an ImageNet backbone, not the eval path).

26. user (2026-07-06T19:36:28.815Z)

<https://github.com/nttcom/WASB-SBDT>.

above issue by employing encoder-decoder CNN models. For example, DeepBall [40, 41] positions. BallSeg [80] is a modification of ICNet [108], so that two consecutive frames can Badminton [75]. This dataset was introduced by the TrackNetV2 work [75]. Following the comparison on various SBDT datasets. In particular, we implemented DeepBall [40, 42], BallSeg [80], TrackNetV2 [75] and MonoTrack [50]. For DeepBall, since its original mediate feature dimension, which is called DeepBall-Large in the followings. Also, we deployed an unpublished variant9 of TrackNetV2, where residual connection and transposed convolution are additionally employed. We call this variant as ResTrackNetV2. accuracy of our TrackNetV2 [75] implementation on the Badminton dataset is 85.6, while 7<https://nol.cs.nctu.edu.tw:234/open-source/TrackNetV2>

DeepBall [40, 41]	0.1M	44.5	92.7	26.3	44.6	47.4	32.3	47.0	52.1	52.4	38.6	60.0	57.1	64.4	50.7	49.2
21.10.0	12.9	0.0	30.3	DeepBall-Large	1.0M	44.9	89.5	34.0	42.0	46.7	31.6	35.1	47.7	50.6	36.8	59.5
53.0	70.4	57.5	56.5	21.1	57.2	47.5	36.6	30.9	BallSeg [80]	12.7M	36.1	92.6	20.0	64.8	71.7	57.5
56.8	62.7	77.9	9.72.2	68.4	75.0	19.5	17.5	8.5	18.2	16.8	20.5	5.3	29.5	TrackNetV2 [75]	11.3M	86.6
97.7	77.2	66.0	89.4	81.4	80.6	55.3	90.5	85.6	83.6	77.0	83.6	73.8	72.3	17.6	78.8	69.3
64.6	28.0	ResTrackNetV2	1.2M	84.6	97.4	75.5	56.2	90.3	82.8	81.7	59.0	89.4	84.0	82.2	71.3	84.2
74.7	74.7	28.6	77.9	68.2	66.0	38.2	MonoTrack [50]	2.9M	85.2	97.4	78.6	58.0	92.1	85.9	87.3	64.1
90.9	85.9	84.9	75.5	85.1	75.9	72.1	19.7	80.8	71.3	65.3	32.1					

correctness (or, superiority) of our TrackNetV2 implementation. Unfortunately, such a validation cannot be performed for the remaining five methods: The original DeepBall [41] was makes the validation intractable. For BallSeg [80], neither its specific architecture is presented nor the benchmark is publicly available. The MonoTrack paper [50] does not explain their experimental protocol at all, and the remaining two (DeepBall-Large and ResTrack- Following TrackNetV2 and its variants [50, 75], N (cf. §3.1) is set to 3 and each image is DeepBall

DeepBall-Large

BallSeg

TrackNetV2

ResTrackNetV2

MonoTrack

WASB (Step=3)

WASB (Step=1)

thresholds. Interestingly, the performances of DeepBall [40, 42] and BallSeg [80] heavily depend on the dataset, while TrackNetV2 [75], ResTrackNetV2 and MonoTrack [50] stably [41] Jacek Komorowski., Grzegorz Kurzejamski., and Grzegorz Sarwas. DeepBall: Deep [50] Paul Liu and Jui-Hsien Wang. MonoTrack: Shuttle Trajectory Reconstruction From Chung, and Tsi-Uí ?Ik. TrackNetV2: Efficient Shuttlecock Tracking Network. In 2020

27. user (2026-07-06T19:36:30.645Z)

```

==== hrnet.py init_weights context (torch.load at ~495) ====
    x = self.deconv_layers[i][scale](x)
    y = self.final_layers[scale](x)
    y_out[scale] = y
    return y_out

def init_weights(self, pretrained='',):
    logger.info('=> init weights from normal distribution')
    for m in self.modules():
        if isinstance(m, nn.Conv2d):
            nn.init.kaiming_normal_(
                m.weight, mode='fan_out', nonlinearity='relu')
        elif isinstance(m, nn.BatchNorm2d):
            nn.init.constant_(m.weight, 1)
            nn.init.constant_(m.bias, 0)
    if os.path.isfile(pretrained):
        pretrained_dict = torch.load(pretrained)
        logger.info('=> loading pretrained model {}'.format(pretrained))
        model_dict = self.state_dict()
        pretrained_dict = {k: v for k, v in pretrained_dict.items()
                           if k in model_dict.keys()}
        for k, _ in pretrained_dict.items():
            logger.info(
                '=> loading {} pretrained model {}'.format(k, pretrained))
        model_dict.update(pretrained_dict)
        self.load_state_dict(model_dict)

"""
if __name__=='__main__':

    import yaml
    from attrdict import AttrDict
    import torch

```

28. assistant (2026-07-06T19:37:32.533Z)

I now have all the primary-source facts I need. Let me parse the badminton column of Table 2 to isolate the exact WASB vs TrackNetV2 badminton numbers.

Reading Table 2 by column groups (each method row is: #param, then Soccer[F1 Acc AP FPS], Tennis[F1 Acc AP FPS], **Badminton[F1 Acc AP FPS]**, Volleyball[...], Basketball[...]):

```

- **TrackNetV2 [75]** `11.3M | Soccer: 86.6 97.7 77.2 66.0 | Tennis: 89.4 81.4 80.6 55.3 |
Badminton: 90.5 85.6 83.6 77.0 | ...` ? Badminton **F1=90.5, Acc=85.6, AP=83.6**
- **WASB (Ours, Step=3)** `1.5M | Soccer: 88.3 97.9 83.6 55.7 | Tennis: 94.0 89.0 91.0 58.2 |
Badminton: 91.6 87.0 88.5 70.4 | ...` ? Badminton **F1=91.6, Acc=87.0, AP=88.5**
- **WASB (Ours, Step=1)** `1.5M | ... Tennis: 95.6 91.8 94.2 35.2 | Badminton: 93.1 89.0 91.6
34.3 | ...` ? Badminton **F1=93.1, Acc=89.0, AP=91.6**

```

All facts are corroborated from primary sources. Here is my report.

WASB-SBDT Research Findings

All facts below are from primary sources (the repo files themselves via GitHub API, and the BMVC 2023 paper PDF).

1. Repository code license ? MIT (permissive, commercial use OK)

```

- The repo is **MIT-licensed**. GitHub API reports `spx_id: MIT`. The `LICENSE.md` file text:
**"MIT License ? Copyright (c) 2023 NTT Communications Corporation ... Permission is hereby
granted, free of charge, ... to use, copy, modify, merge, publish, distribute, sublicense,
and/or sell copies of the Software..."

```

- Source: ``https://github.com/nttcom/WASB-SBDT/blob/main/LICENSE.md`` and the README license badge (``License: MIT``).
- ****The MIT license covers the code only.**** It does not, by itself, license the pretrained weights or the third-party training datasets (see item 2).

2. Pretrained weights & badminton training data ? LICENSE GAP (open question, matters for commercial use)

- ****Weights:**** ``wasb_badminton_best.pth.tar`` is one of five WASB checkpoints (one per sport). They're distributed as ``.pth.tar`` files via ****Google Drive**** links in ``MODEL_ZOO.md``. The Model Zoo also hosts weights for 6 re-implemented baselines (DeepBall, DeepBall-Large, BallSeg, TrackNetV2, ResTrackNetV2, MonoTrack). Source: ``https://github.com/nttcom/WASB-SBDT/blob/main/MODEL_ZOO.md``.
- ****Badminton training dataset = the TrackNetV2 badminton dataset (NCTU).**** ``GET_STARTED.md`` instructs downloading Badminton from ``https://nol.cs.nctu.edu.tw:234/open-source/TrackNetV2`` (``TrackNetV2.zip``). The paper confirms: ****"Badminton [75]. This dataset was introduced by the TrackNetV2 work [75]."** The badminton config (``src/configs/dataset/badminton.yaml``) uses 26 training matches (``match1`?```match26``) and 3 test matches ? matching the TrackNetV2 "Shuttlecock" dataset structure. Sources: ``GET_STARTED.md``, ``src/configs/dataset/badminton.yaml``, BMVC PDF.
- ****OPEN QUESTION / RISK ? no license is stated on the weights or the TrackNetV2 dataset.**** The repo's MIT LICENSE.md makes ****no mention**** of the pretrained weights or datasets, and ``MODEL_ZOO.md/GET_STARTED.md`` attach ****no usage terms****. The WASB weights were trained on NCTU's TrackNetV2 dataset, whose own license terms are not restated in this repo and are not obviously permissive. ****For commercial use you should not assume the MIT code license extends to the badminton weights**** ? those weights are a derivative of the TrackNetV2 academic dataset. This needs independent verification with the TrackNetV2/NCTU dataset terms before commercial deployment. (The WASB paper does not grant a weights license either.)

3. Torch version pin & backbone

- ****Torch is pinned to `torch==1.11.0+cu113`**** (with ``torchvision==0.12.0+cu113``, ``numpy==1.22.4``). There is ****no `requirements.txt`**** in the repo ? dependencies live in the ``Dockerfile``:
 - ``RUN pip3 install numpy==1.22.4 torch==1.11.0+cu113 torchvision==0.12.0+cu113 --extra-index-url https://download.pytorch.org/whl/cu113``
 - ``RUN pip3 install hydra-core==1.2.0 tqdm==4.64.0 opencv-python==4.6.0.66 scikit-learn==1.1.1 scikit-image==0.19.3 pandas==1.3.5 einops==0.4.1 timm==0.6.5 matplotlib==3.5.2``
 - Base image ``nvcr.io/nvidia/tensorrt:21.06-py3``; README says "Tested with Python3.8, CUDA11.3". Source: ``https://github.com/nttcom/WASB-SBDT/blob/main/Dockerfile``.
- ****Yes ? HRNet is the WASB backbone/detector.**** ``src/configs/model/wasb.yaml`` begins with ``name: hrnet`` and specifies the full HRNet multi-branch config (STAGE1?STAGE4, NUM_BRANCHES 1?4). The model code is ``src/models/hrnet.py``. WASB is only ****1.5M params**** (vs TrackNetV2's 11.3M). Sources: ``src/configs/model/wasb.yaml``, ``src/models/hrnet.py``.

4. Torch 1.11 ? 2.x compatibility risks

- ****`torch.load` `weights\_only` ? REAL BREAK RISK.**** ``src/detectors/detector.py`` line 58 calls ``checkpoint = torch.load(model_path)`` with ****no `weights\_only=` argument****, then ``self._model.load_state_dict(checkpoint['model_state_dict'])``. On ****torch ? 2.6**** the default flipped to ``weights_only=True``. Whether this raises depends on what's pickled in the ``.pth.tar``: a plain tensor/state-dict dict loads fine, but if the checkpoint pickled any non-tensor objects (e.g. numpy scalars, argparse/OmegaConf objects, custom classes) it will raise ``UnpicklingError`/`WeightsUnpickler error``. ****Mitigation is trivial:**** pass ``weights_only=False`` (trusted file) or add safe globals. ``src/models/hrnet.py`` line 495 has a second bare ``torch.load(pretrained)`` in ``init_weights()``, but that path is only hit when loading an ImageNet-pretrained backbone during training, not during the eval/inference flow. Source: ``src/detectors/detector.py``, ``src/models/hrnet.py``.
- ****hydra-core / omegaconf pins ? LOW-MODERATE risk.**** The repo pins ``hydra-core==1.2.0`` (which pulls ``omegaconf`` ~2.2). This is an old Hydra; it generally still installs on modern Python but Hydra 1.2 + newer omegaconf/pydantic-free stacks can conflict. Config code uses ``HydraConfig.get().run.dir`` (stable API). No omegaconf version is pinned directly. Recommend upgrading to hydra-core ?1.3 and re-testing config resolution. Source: ``Dockerfile``, ``src/detectors/detector.py``.
- ****Removed/renamed torch APIs ? LOW risk for this HRNet code.**** The HRNet implementation uses only stable ops: ``nn.BatchNorm2d(..., momentum=BN_MOMENTUM)``, ``nn.Conv2d``,

`nn.init.kaiming\_normal\_`, and `nn.DataParallel`. I found **no** use of the classic HRNet footguns (`F.upsample`, `Variable`, `.volatile`, `async`, `.data[...]`). The main friction points are: (a) `nn.DataParallel` still works but is deprecated in favor of DDP; (b) upsampling in HRNet fusion typically needs an explicit `align\_corners=` ? not a hard break, but numerics can shift. Overall the model code should port to torch 2.x with minimal changes; the checkpoint-load default is the one thing most likely to fail out of the box. Source: `src/models/hrnet.py`, `src/detectors/detector.py`.

5. Benchmark & metrics ? badminton numbers (Table 2)

- **Dataset:** the **TrackNetV2 Badminton dataset** (30 fps, 1280×720; 26 train + 3 test matches). **Metrics:** F1, Accuracy (Acc.), and Average Precision (AP), computed at a distance threshold **? = 4 pixels**. Source: BMVC PDF Table 1 & Table 2 caption: **"We set the distance threshold ? = 4 [pixel] to compute F1, Accuracy (Acc.) and Average Precision (AP)."**
- **Badminton results (Table 2), F1 / Acc / AP:**
 - **WASB (Ours, Step=1):** F1 = 93.1, Acc = 89.0, AP = 91.6
 - **WASB (Ours, Step=3):** F1 = 91.6, Acc = 87.0, AP = 88.5
 - TrackNetV2 [75] (their re-impl): F1 = 90.5, Acc = 85.6, AP = 83.6
 - MonoTrack [50]: F1 = 90.9, Acc = 85.9, AP = 84.9
 - ResTrackNetV2: F1 = 89.4, Acc = 84.0, AP = 82.2
 - BallSeg: F1 = 79.9, Acc = 72.2, AP = 68.4
- **SOTA claim vs TrackNetV2 ? YES.** WASB beats TrackNetV2 on badminton on every metric (F1 93.1 vs 90.5; Acc 89.0 vs 85.6; AP 91.6 vs 83.6) with far fewer params (1.5M vs 11.3M). The paper claims WASB is **"substantially superior to existing methods on all the sports categories"** and beats 6 SOTA methods across 5 datasets. The paper also validates its own re-implementation: **"the accuracy of our TrackNetV2 implementation on the Badminton dataset is 85.6, while Table IV in [75] shows that the original implementation scores 85.2"** ? i.e., their TrackNetV2 baseline is faithful. Source: BMVC PDF Table 2 and §Experiments.

6. Paper metadata

- **Title:** **"Widely Applicable Strong Baseline for Sports Ball Detection and Tracking"**
- **Authors:** Shuhei Tarashima, Muhammad Abdul Haq, Yushan Wang, Norio Tagawa (NTT Communications; Tokyo Metropolitan University co-authors)
- **Venue / Year:** **"BMVC 2023"** (34th British Machine Vision Conference, Aberdeen, UK). Note: it is **"BMVC 2023, not ECCV"** ? the "ECCV 2022/2023" attribution in the prompt is incorrect.
- **arXiv:** **"2311.05237"** (preprint released 11/09/2023). BMVC proceedings: **"https://proceedings.bmvc2023.org/310/"**; PDF: **"https://papers.bmvc2023.org/0310.pdf"**.
- **BibTeX (from README):** **"@inproceedings{tarashima2023wasb, title={Widely Applicable Strong Baseline for Sports Ball Detection and Tracking}, author={Tarashima, Shuhei and Haq, Muhammad Abdul and Wang, Yushan and Tagawa, Norio}, booktitle={BMVC}, year={2023}"**

Sources

- Repo: <https://github.com/nttcom/WASB-SBDT>
- LICENSE.md: <https://github.com/nttcom/WASB-SBDT/blob/main/LICENSE.md>
- MODEL_ZOO.md: https://github.com/nttcom/WASB-SBDT/blob/main/MODEL_ZOO.md
- GET_STARTED.md: https://github.com/nttcom/WASB-SBDT/blob/main/GET_STARTED.md
- Dockerfile (dependency pins): <https://github.com/nttcom/WASB-SBDT/blob/main/Dockerfile>
- Checkpoint loader (`torch.load`): `src/detectors/detector.py`; HRNet backbone: `src/models/hrnet.py`; WASB/badminton configs: `src/configs/model/wasb.yaml`, `src/configs/dataset/badminton.yaml`
- arXiv abstract: <https://arxiv.org/abs/2311.05237>
- BMVC 2023 paper PDF (Table 2 numbers): <https://papers.bmvc2023.org/0310.pdf> ; proceedings: <https://proceedings.bmvc2023.org/310/>
- TrackNetV2 badminton dataset origin: <https://nol.cs.nctu.edu.tw:234/open-source/TrackNetv2>

Top open questions to resolve before commercial use

1. **Weights/dataset license (highest priority):** The MIT code license does **not** clearly cover the badminton weights, which were trained on NCTU's TrackNetV2 dataset. Verify the TrackNetV2 dataset's own terms and whether redistributing/using WASB badminton weights commercially is permitted. NTT has not attached explicit weight terms.

2. `**.pth.tar` contents:**` Confirm what is pickled inside ``wasb_badminton_best.pth.tar`` to know whether ``weights_only=True`` (torch ?2.6) will load it or need ``weights_only=False`/safe-globals`.
3. `**Third-party baseline weights**` (TrackNetV2, MonoTrack, etc.) in the Model Zoo are re-implementations by NTT but derive from other papers' methods/datasets ? same license-provenance caveat applies if you use those rather than WASB.